3D WEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

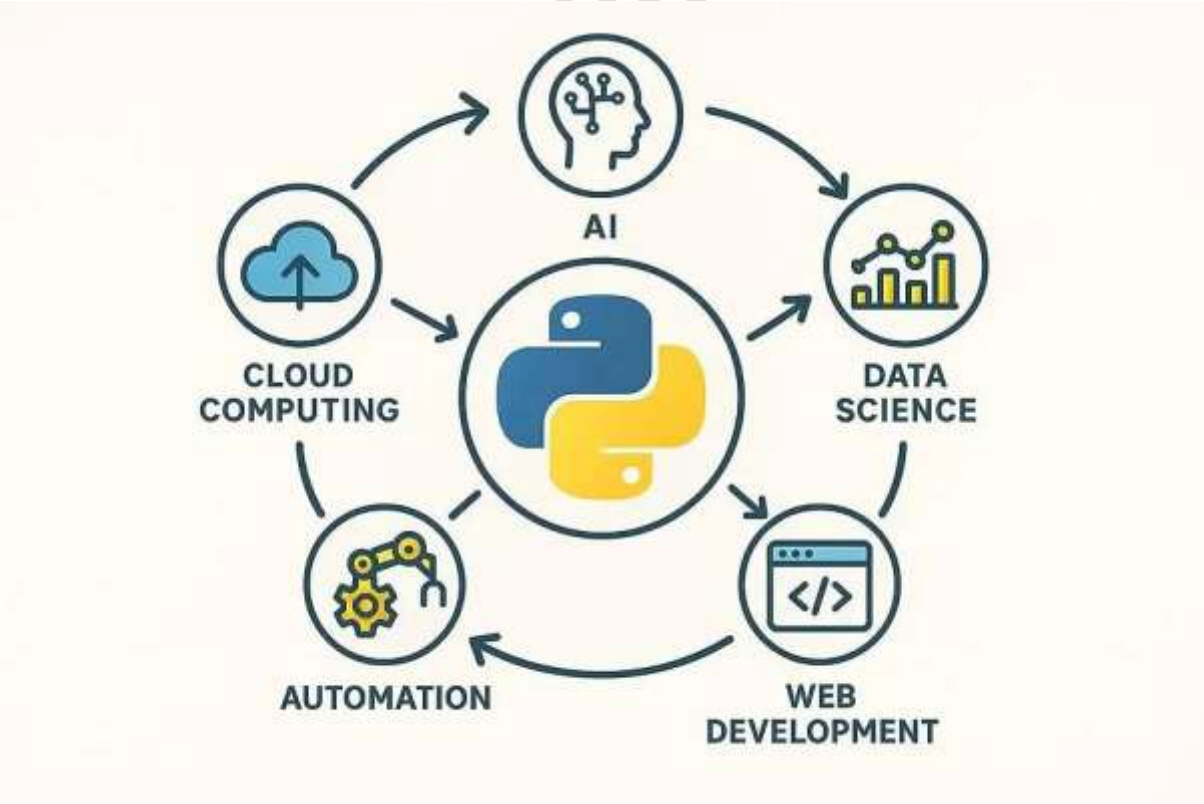
Daily Teaching : 2 Hours

Total Classes : 100

FOUNDATION

Level 4 – AI Professional Foundation(11th–12th Standard)

Class 1 – Advanced Python Introduction



Learning Objectives

By the end of this class, students will be able to:

- Understand what Python is and why it is widely used.
- Explain the difference between basic and advanced Python programming.
- Understand Python's role in AI, Machine Learning, Data Science and automation.
- Set up Python on a computer.
- Use Python IDLE/VS Code/Jupyter Notebook.
- Run their first Python program.
- Revise important Python fundamentals.
- Identify common Python errors and fix simple problems.

3DWEBSOFT FOUNDATION

1. What is Python?

Python is a high-level, general-purpose programming language used to create software, websites, automation tools, data-analysis applications, AI systems and many other applications.

Python was designed to have a **simple and readable syntax**, which makes it suitable for beginners while also being powerful enough for professional development.

Simple Example

```
name = "John"

print("Hello", name)
```

Output

```
Hello John
```

Python allows us to give instructions to a computer using programming statements.

2. Why Learn Advanced Python?

Students may already know basic Python concepts such as:

- Variables
- Data types
- Operators
- if statements
- Loops
- Functions
- Lists
- Dictionaries

Advanced Python goes further and teaches students how to write **larger, cleaner and more reusable programs**.

Advanced Python includes:

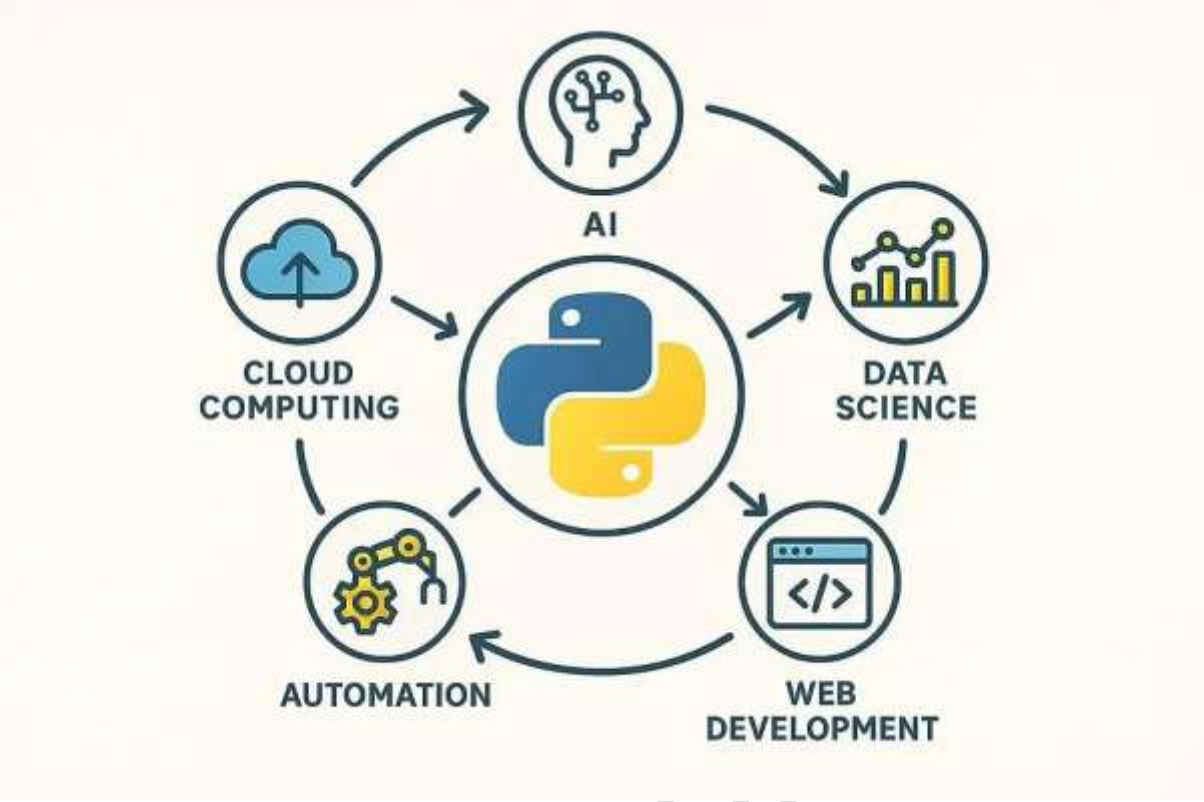
- Advanced functions
- Lambda functions
- List/dictionary comprehensions
- Modules and packages
- File handling
- JSON and CSV processing
- Exception handling
- Object-Oriented Programming
- Iterators and generators
- Decorators
- APIs

3DWEBSOFT FOUNDATION

- Working with libraries
- Data processing
- Python for AI and ML

3. Where is Python Used?





Python is used in many areas:

Artificial Intelligence

Python is widely used for developing AI applications.

Machine Learning

Students can use Python libraries to train machine-learning models.

Data Science

Python can process and visualize large amounts of data.

Web Development

Python can be used to build web applications.

Automation

Python can automate repetitive tasks.

Cybersecurity

3DWEBSOFT FOU

Python can be used for security analysis, defensive tools and authorized security testing.

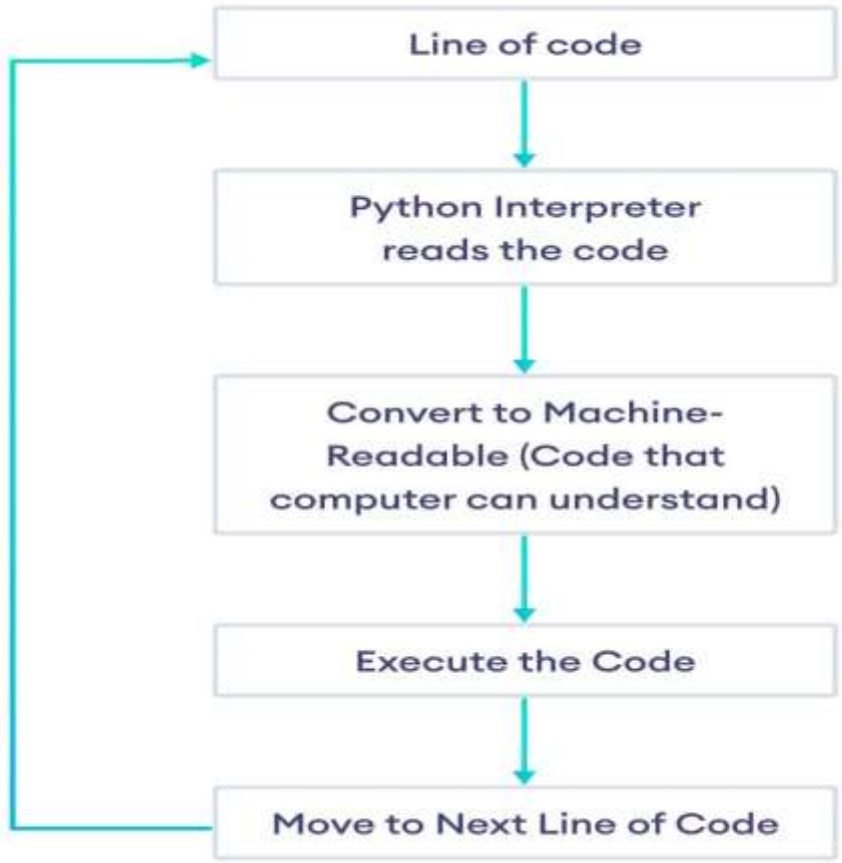
Game Development

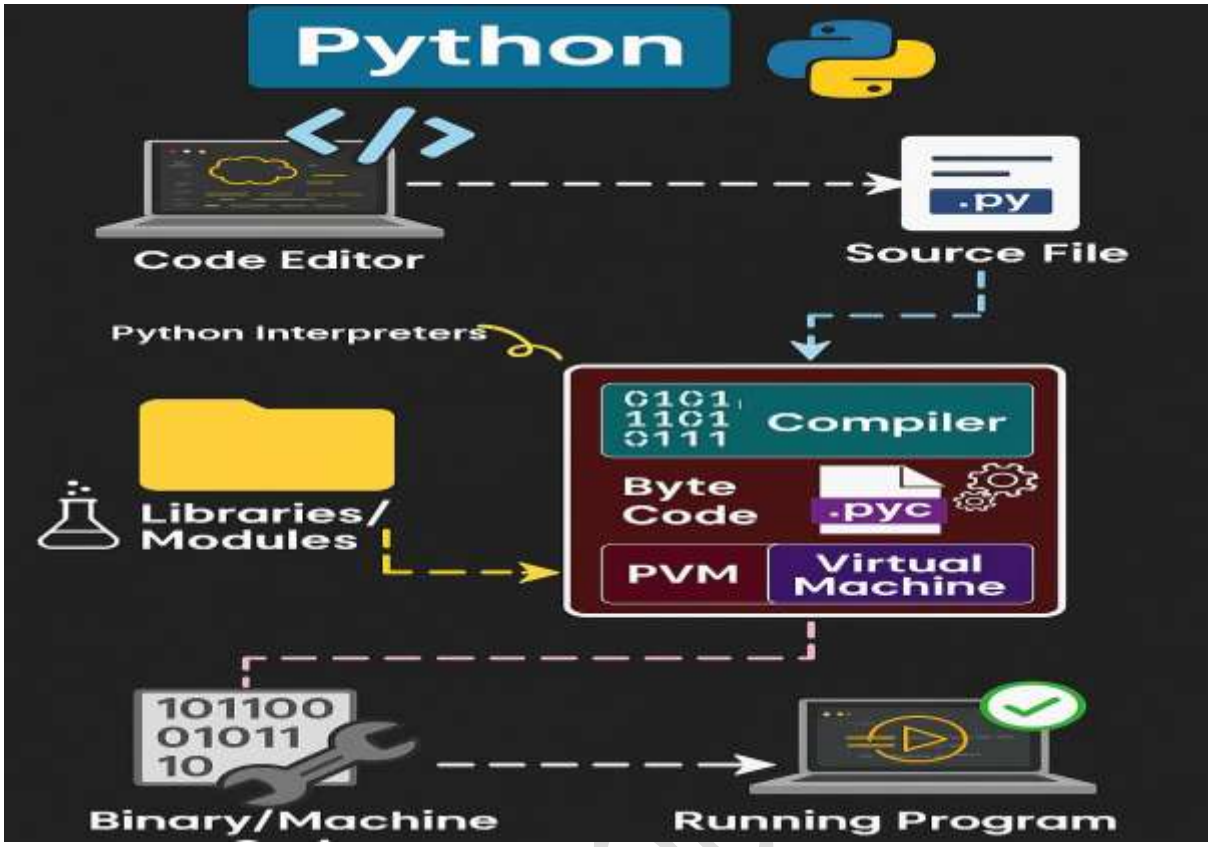
Python can be used to create simple games and simulations.

4. Python Program Execution

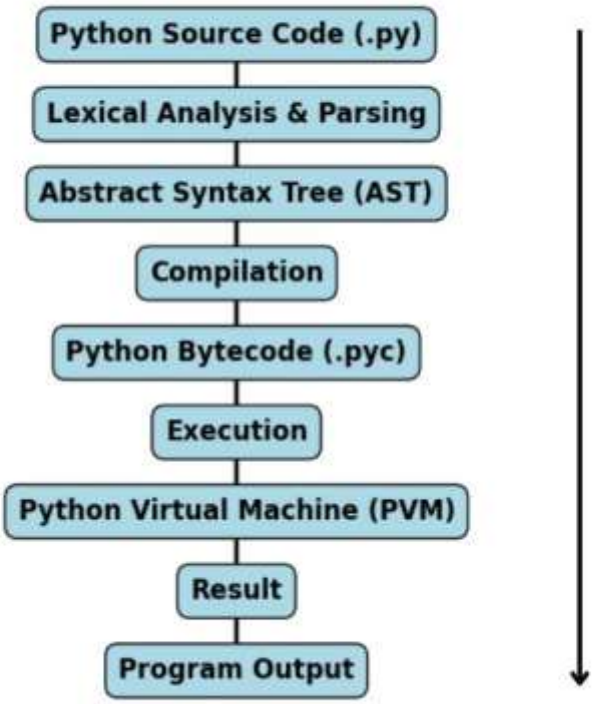
A simple way to understand Python execution:

Student writes code → Python interpreter reads code → Python executes instructions → Output appears





Python Execution Flow Overview



Example

```
a = 10

b = 20


result = a + b


print(result)
```

Output

```
30
```

The Python interpreter executes the statements and produces the result.

5. Python Environment Setup

Before writing advanced programs, students need a Python programming environment.

There are several options:

1. **Python IDLE**
2. **VS Code**
3. **Jupyter Notebook**
4. Other Python-compatible development environments

For this course, **VS Code + Python** is a good setup for project-based learning.

6. Installing Python

Step 1 – Download Python

Students can download Python from the official Python website.

Official Python Website

3DWEBSOFT FOUNDATION

Step 2 – Install Python

During installation on Windows, make sure the option to **add Python to PATH** is selected.

Then select the installation option.

Step 3 – Verify Installation

Open **Command Prompt**.

Type:

python --version

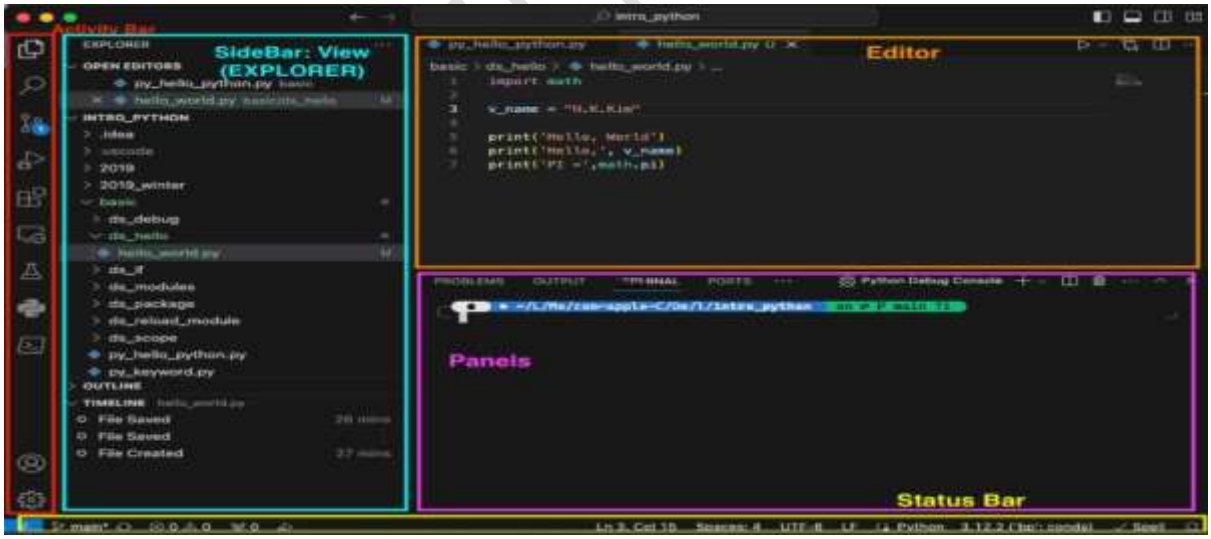
or, depending on the system:

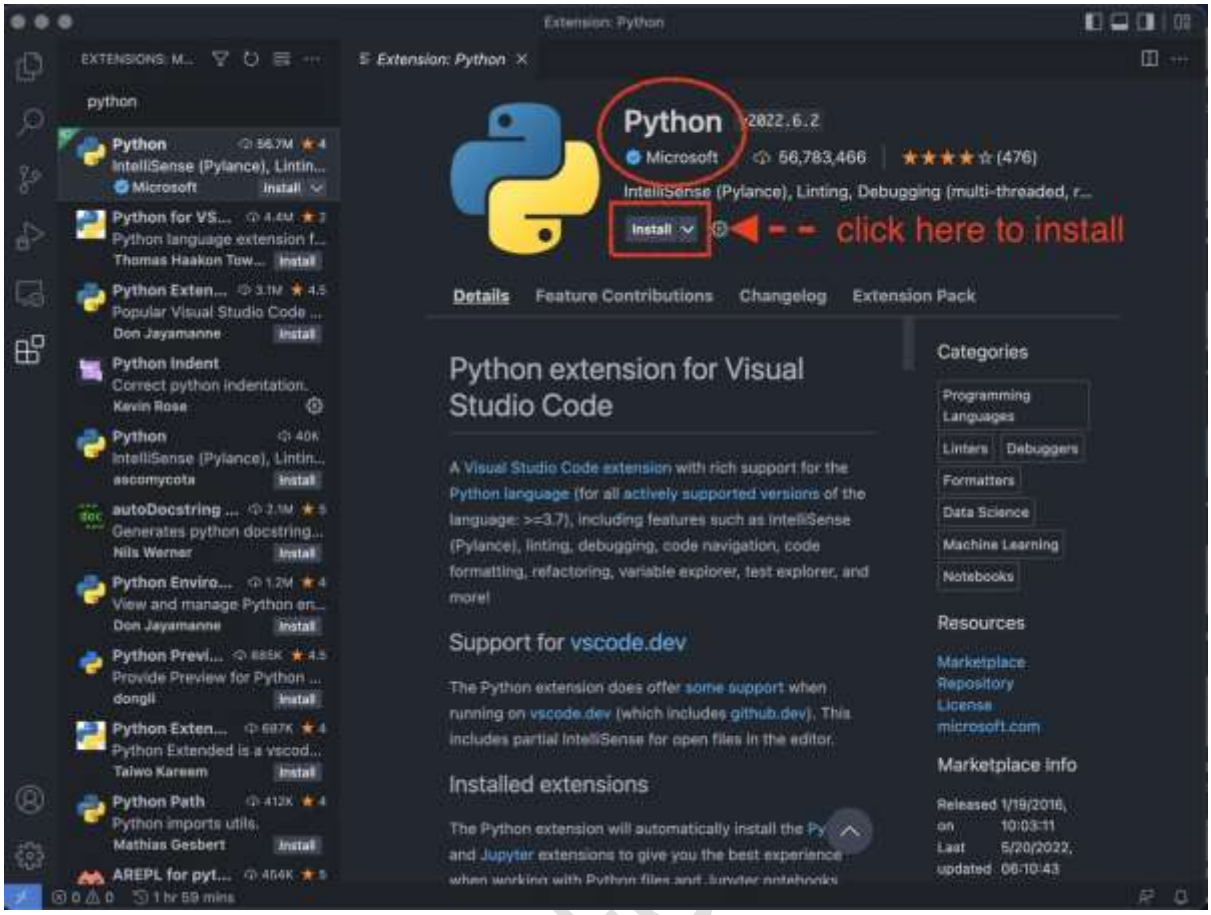
py --version

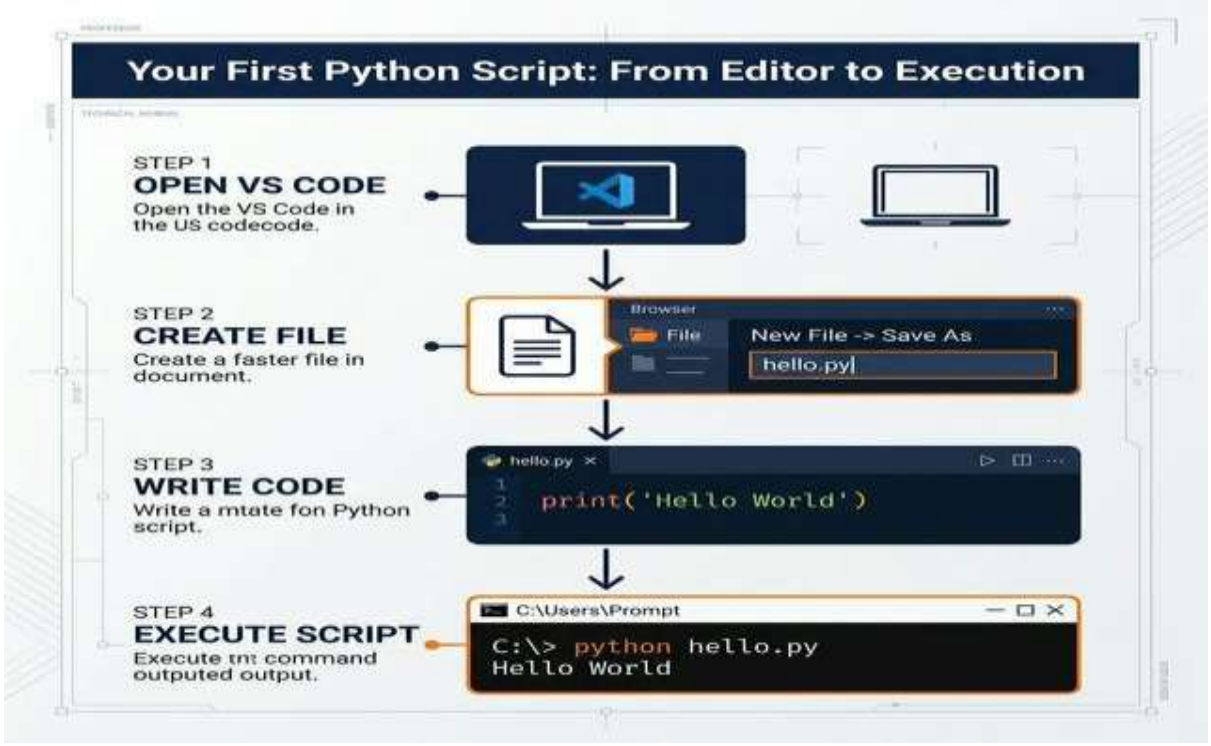
A Python version number should appear.

T FOUNDATION

7. VS Code Setup




Extension: Python
EXTENSIONS: M...
python
Extension: Python
Python
Microsoft
Install
Python for VS...
Python language extension f...
Thomas Haakon Tow...
Python Exten...
Popular Visual Studio Code ...
Don Jayamanne
Python Indent
Correct python indentation.
Kevin Rose
Python
IntelliSense (Pylance), Lintin...
ascomycota
autoDocstring ...
Generates python docstring...
Nils Werner
Python Enviro...
View and manage Python en...
Don Jayamanne
Python Previ...
Provide Preview for Python ...
dongli
Python Exten...
Python Extended is a vscod...
Taiwo Kareem
Python Path
Python imports utils.
Mathias Gesbert
AREPL for pyt...
1 hr 59 mins
Python
2022.6.2
Microsoft
56,783,466
★★★★☆(476)
IntelliSense (Pylance), Linting, Debugging (multi-threaded, r...
Install
click here to install
Details
Feature Contributions
Changelog
Extension Pack
Python extension for Visual Studio Code
A Visual Studio Code extension with rich support for the Python language (for all actively supported versions of the language: >=3.7), including features such as IntelliSense (Pylance), linting, debugging, code navigation, code formatting, refactoring, variable explorer, test explorer, and more!
Support for vscode.dev
The Python extension does offer some support when running on vscode.dev (which includes github.dev). This includes partial IntelliSense for open files in the editor.
Installed extensions
The Python extension will automatically install the Py and Jupyter extensions to give you the best experience when working with Python files and Jupyter notebooks
Categories
Programming Languages
Linters
Debuggers
Formatters
Data Science
Machine Learning
Notebooks
Resources
Marketplace
Repository
License
microsoft.com
Marketplace info
Released on 1/19/2016, 10:03:11
Last updated 5/20/2022, 06:10:43


Your First Python Script: From Editor to Execution
STEP 1
OPEN VS CODE
Open the VS Code in the US codecode.
STEP 2
CREATE FILE
Create a faster file in document.
Browser
File
New File -> Save As
hello.py
STEP 3
WRITE CODE
Write a mtate fon Python script.
hello.py
1
2
3
print('Hello World')
STEP 4
EXECUTE SCRIPT
Execute tnt command outputed output.
C:\Users\Prompt
C:\> python hello.py
Hello World

VS Code is a source-code editor that can be configured for Python development.

Basic setup

1. Install VS Code.
2. Open VS Code.
3. Open the Extensions section.
4. Search for **Python**.
5. Install the Python extension from Microsoft.
6. Create a folder for the course.
7. Create a file named:

hello.py

8. First Python Program

Type:

print("Hello, World!")

Run the program.

Expected Output

Hello, World!

Student Challenge

Change the program to:

print("My name is _____")

print("I am studying in Class _____")

print("I am learning Advanced Python")

9. Python Revision

Before moving into advanced concepts, revise the following.

3DWEBSOFT FOUNDATION

Variables

A variable stores a value.

```
name = "Rahul"
age = 17
marks = 92.5
```

Data Types

Common Python data types include:

```
name = "Anita"      # str
age = 17            # int
percentage = 91.5   # float
passed = True       # bool
```

Students should understand that different values can have different data types.

10. Operators Revision

Arithmetic Operators

```
a = 10
b = 3

print(a + b)
print(a - b)
print(a * b)
print(a / b)
print(a % b)
```

Comparison Operators

3DWEBSOFT FOUNDATION

```
a = 10
b = 20

print(a > b)
print(a < b)
print(a == b)
```

Logical Operators

```
age = 17
marks = 85

print(age >= 16 and marks >= 50)
```

11. Conditional Statements

Python uses if, elif and else for decision-making.

```
marks = 85

if marks >= 90:
    print("Excellent")
elif marks >= 75:
    print("Very Good")
else:
    print("Keep Practicing")
```

Real-world example

A school application can use conditions to determine whether a student passed an examination.

3DWEBSOFT FOUNDATION

12. Loops Revision

Loops allow us to repeat instructions.

for loop

```
for i in range(1, 6):
    print(i)
```

Output:

1

2

3

4

5

while loop

```
count = 1

while count <= 5:
    print(count)
    count += 1
```

13. Collections Revision

List

```
students = ["Asha", "Rahul", "Priya"]

print(students)
```

Dictionary

3DWEBSOFT FOUNDATION

```
student = {
   "name": "Asha",
   "age": 17,
   "marks": 90
}

print(student["name"])
```

These concepts will become important when students work with larger datasets and applications.

14. Functions Revision

A function is a reusable block of code.

```
def greet(name):
   print("Hello", name)

greet("Asha")
greet("Rahul")
```

Why functions are useful

Functions help us:

- Avoid repeated code.
- Organize programs.
- Make programs easier to understand.
- Reuse logic.
- Test individual parts of a program.

3DWEBSOFT FOUNDATION

Practical Session

15. Practical Activity 1 – Student Information Program

Ask students to create:

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
school = input("Enter your school name: ")

print("\nStudent Information")
print("Name:", name)
print("Age:", age)
print("School:", school)
```

Expected Result

The program asks the student for information and displays it.

16. Practical Activity 2 – Marks Analyzer

```
marks = float(input("Enter your marks: "))

if marks >= 90:
    print("Grade: A")
elif marks >= 75:
    print("Grade: B")
elif marks >= 60:
    print("Grade: C")
elif marks >= 40:
```

3DWEBSOFT FOUNDATION

```
    print("Grade: D")
else:
    print("Grade: F")
```

Challenge

Modify the program to display:

- Excellent
- Very Good
- Good
- Pass
- Fail

17. Practical Activity 3 – Simple Calculator

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
```

Advanced Challenge

Ask students to add:

- Modulus
- Floor division
- Power

Example:

3DWEBSOFT FOUNDATION

```
print("Remainder:", a % b)

print("Power:", a ** b)
```

18. Common Python Errors

Students should learn that errors are a normal part of programming.

Syntax Error

Example:

```
print("Hello"
```

The closing parenthesis is missing.

Name Error

```
print(student_name)
```

If student_name was never defined, Python reports an error.

Type Error

```
age = "17"

print(age + 5)
```

Here, a string and integer are being combined incorrectly.

ZeroDivisionError

```
result = 10 / 0
```

A number cannot be divided by zero.

19. Interactive Classroom Activity – "Find the Bug"

Give students this program:

```
name = input("Enter your name: ")

age = input("Enter your age: ")
```

3DWEBSOFT FOUNDATION

```
if age >= 18:
    print(name, "is an adult")
else:
    print(name, "is a minor")
```

Question

What is wrong with this program?

Hint

input() normally returns text.

Students should identify that age needs to be converted into an integer:

```
age = int(input("Enter your age: "))
```

Classroom Quiz

Q1. What is Python?

A. Operating system
B. Programming language
C. Database
D. Web browser

Answer: B

Q2. Which function displays output?

A. input()
B. display()
C. print()
D. show()

Answer: C

Q3. Which keyword is used to define a function?

A. function
B. def

3DWEBSOFT FOUNDATION

C. fun
D. define

Answer: B

Q4. Which data type stores True or False?

A. str
B. int
C. bool
D. float

Answer: C

Q5. Which statement is used for decision-making?

A. if
B. loop
C. repeat
D. define

Answer: A

Mini Challenge

"Student Result Analyzer"

Create a Python program that:

1. Takes student name.
2. Takes marks in 3 subjects.
3. Calculates total.
4. Calculates average.
5. Displays pass/fail.
6. Displays a grade.

Example

Enter student name: Aisha

3DWEBSOFT FOUNDATION

Enter Python marks: 90

Enter AI marks: 85

Enter Maths marks: 80

Total: 255

Average: 85

Grade: A

Result: PASS

Homework

Task 1

Write a program to display your:

- Name
- Age
- School
- Class
- Favorite subject
- Career goal

Task 2

Create a calculator that performs five arithmetic operations.

Task 3

Create a program that checks whether a number is:

- Positive
- Negative
- Zero

3DWEBSOFT FOUNDATION

Task 4 – Challenge

Create a program that asks for a student's marks and automatically assigns a grade.

Student Takeaway

Python → Advanced Python → Data → Machine Learning → AI → Generative AI → Cybersecurity → Capstone Project

This first class establishes the foundation for the next classes, where students can move from basic Python into **advanced functions, comprehensions, modules, file handling, OOP, data processing and Python-based AI/ML projects**.

3DWEBSOFT FOUNDATION